THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A black coating forms on copper heated in air. Copper has been: 1

(a) oxidised **(b)** neutralised **(c)** reduced **(d)** decomposed

2. Acids taste: 1

(a) sour **(b)** salty **(c)** bitter **(d)** sweet

3. Non-metallic oxides are generally: 1

(a) acidic **(b)** basic **(c)** neutral **(d)** amphoteric

4. Ionic compounds conduct electricity: 1

(a) in solid state **(b)** only as gases

(c) never **(d)** in molten state and aqueous solution

5. Metals produce a ringing sound when struck. This property is called: 1

(a) lustre **(b)** ductility **(c)** malleability **(d)** sonority

6. A student states: “Unsaturated fats are healthier than saturated fats.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) Both are equally harmful **(d)** Cannot be determined

7. The molecular formula of benzene is: 1

(a) C_5H_{10} **(b)** C_6H_6 **(c)** C_6H_{14} **(d)** C_6H_{12}

8. Rings of cartilage are present in the: 1

(a) alveoli **(b)** trachea **(c)** stomach **(d)** oesophagus

9. Dwarfism results from deficiency of: 1

(a) thyroxine **(b)** growth hormone **(c)** adrenaline **(d)** insulin

10. The part of the flower that develops into a fruit is the: 1

(a) stigma **(b)** ovary **(c)** ovule **(d)** anther

11. The pollen tube reaches the ovule through the: 1

(a) sepal **(b)** filament **(c)** style **(d)** petal

12. The traits that are transmitted from parents to offspring are called: 1

(a) inherited traits **(b)** mutant traits **(c)** acquired traits **(d)** learnt traits

13. For pea plants, Student A says round seed is dominant over wrinkled; Student B says wrinkled is dominant. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

14. The focal length of a convex lens is 25 cm. Its power is: 1

(a) +25 D **(b)** −4 D **(c)** +4 D **(d)** +0.25 D

15. A prism has how many rectangular lateral surfaces? 1

(a) five **(b)** three **(c)** two **(d)** four

16. The direction of conventional current is: 1

(a) from negative to positive terminal outside
(b) same as electron flow
(c) opposite to the flow of electrons
(d) random

17. The unit of electric power is: 1

(a) joule **(b)** ohm **(c)** watt **(d)** volt

18. The pattern of field lines inside a solenoid shows that the field is: 1

(a) strongest at the ends **(b)** zero
(c) the same at all points **(d)** circular

19. The chemical responsible for ozone depletion is: 1

(a) methane **(b)** nitrogen
(c) carbon monoxide **(d)** chlorofluorocarbon

20. Kulhads (disposable clay cups) were not adopted on a large scale because: 1

(a) they are non-biodegradable **(b)** they are toxic
(c) they are costly to import **(d)** it would lead to loss of fertile topsoil

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What happens when quicklime is added to water? Write the equation. 2

22. What are alkalis? Give two examples. 2

23. A student says capillaries have thick walls. Justify or refute. 2

24. What is the role of the receptor? Name the receptors for taste. 2

25. What is the difference between a gamete and a zygote? 2

26. What are the two kinds of sperm produced by a human male? 2

27. What is the magnification of an image that is twice the size of the object and inverted? 2

28. Name two ways of increasing the strength of an electromagnet. 2

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SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Write balanced equations: **3**
(i) potassium reacts with water
(ii) zinc carbonate is heated
(iii) carbon dioxide is passed through lime water.

30. Write two differences between metals and non-metals on the basis of chemical properties. **3**

31. State the functions of the pituitary gland. Why is it called the master gland? **3**

32. Describe multiple fission in Plasmodium. **3**

33. State the causes and correction of presbyopia. **3**

34. How can you show that the magnetic field due to a solenoid is like that of a bar magnet? **3**

35. How is washing soda obtained from baking soda? Write the equations. **3**

OR

How does a non-metallic oxide react with a base? Write the equation for carbon dioxide and calcium hydroxide, and conclude the nature of the oxide. **3**

36. What is the importance of a natural ecosystem like a forest? (three) **3**

OR

How are UV rays harmful and how does ozone protect us? **3**

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Draw (describe) the ray diagrams for a convex mirror for an object at infinity and at a finite distance. **5**
(ii) Why is a convex mirror called a diverging mirror?
(iii) Give one use.

OR

(i) State the laws of refraction. **5**
(ii) Light passes from air into a medium where its speed is 1.5×10^8 m/s. Find the refractive index.
(iii) Explain why a tank of water appears shallower than it actually is.

38. **Case study:** Food is cooked in a gas stove. The bottom of a vessel sometimes becomes black. **5**
(i) Why does this happen?
(ii) What does it indicate about the air holes?
(iii) What fuel is used in LPG?
(iv) Why is incomplete combustion dangerous?
(v) Write the equation of complete combustion of butane.

OR

Case study: A student mixed 1 mL ethanol and 1 mL glacial acetic acid with a few drops of conc. H_2SO_4 and warmed it in a water bath. A sweet smell was observed. 5

(i) Name the reaction.

(ii) Write the equation.

(iii) Role of conc. H_2SO_4?

(iv) What happens when the product is treated with NaOH?

(v) Give one use of the product.

39. (i) 100 J of heat is produced each second in a 4 Ω resistor. Find the potential difference across it. 5

(ii) A 220 V, 1 kW kettle is used on a 110 V supply. Find the power consumed (assume constant resistance).

OR

(i) Derive $R = \rho l / A$ from the dependence of resistance. 5

(ii) Find the length of a nichrome wire of resistance 10 Ω with area 2×10^{-7} m² ($\rho = 1 \times 10^{-6}$ Ω m).

40. (i) Describe the mechanism of breathing in humans. 5

(ii) Why is the residual volume of air in the lungs important?

OR

Case study: A kidney patient undergoes haemodialysis twice a week. 5

(i) What is the principle of dialysis?

(ii) What is removed from the blood?

(iii) How is a nephron similar to an artificial kidney?

(iv) How does it differ?

(v) Name the main waste removed.

END OF QUESTION PAPER **Total: 80 Marks**

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